Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2007-I-Q04] | Part A (Q1-Q5)

Answer: 105

1. Collinearity allows either side of the star, so relative angular displacement must be a half-integer number of turns. The successive relative rates are both $1/60 - 1/84 = 1/84 - 1/140 = 1/210$.

2. Thus the first positive time making relative displacement a multiple of $1/2$ is $210/2 = 105$ years; the third pair then differs by one full turn.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2010-I-Q02] | Part A (Q1-Q5)

Answer: 109

1. For every $j \geq 3$, the factor is -1 modulo 1000. There are 997 such factors.

2. The product is therefore $9 \cdot 99(-1)^{997} = -891 \equiv 109 \pmod{1000}$.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2025-I-Q05] | Part A (Q1-Q5)

Answer: 279

1. Digit sum is 36. The divisibility-by-11 criterion forces each parity class of four positions to have digit sum 18: the alternating difference lies between -16 and 16 and is even, so its only possible multiple of 11 is zero.

2. There are eight four-subsets summing to 18: $1278, 1368, 1458, 1467, 2358, 2367, 2457, 3456$. Across these subsets, the total number of even-digit incidences is 16, since complements pair to all four even digits.

3. Choose the even-position subset, its even units digit, and then order the remaining positions: $N = 16 \cdot 3! \cdot 4! = 2304$. Thus $N - 2025 = 279$.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-I-Q09] | Part B (Q6-Q10)

Answer: 757

1. $h = 1, 2$ cause overlap, so $h \geq 3$. If $h \geq 5$, picket 4 forces $t = 2$: the third painter could reach it only with $u = 1$, which would itself cause overlaps. Then picket 5 forces $u = 2$, after which the second and third painters cover every picket from 2 on, overlapping the first painter's next picket. Thus $h = 3$ or 4.

2. For $h = 3$, picket 5 forces $t = 3$, because $u = 2$ would overlap at 7; the remaining residue class forces $u = 3$. This gives 333.

3. For $h = 4$, picket 4 forces $t = 2$ and the remaining class $3 \pmod 4$ forces $u = 4$, giving 424. These two partitions work, and their sum is 757.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-II-Q07] | Part B (Q6-Q10)

Answer: 112

1. The square-sum formula requires $1200 \mid k(k+1)(2k+1)$. A factor 3 is automatic. All factors of 2 lie in the even one of $k, k+1$, requiring $k \equiv 0, 15 \pmod{16}$.

2. For the prime 5, the three factors are pairwise nonoverlapping modulo 5, so 25 must divide one of them. Thus $k \equiv 0, 24, 12 \pmod{25}$.

3. Combining the six residue pairs gives least positive representatives $112, 175, 224, 287, 399, 400$. The least is 112, which also verifies the original square-sum condition.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2011-I-Q07] | Part B (Q6-Q10)

Answer: 016

1. $m = 1$ is impossible. For $m \geq 2$, reduction modulo $m - 1$ forces $m - 1 \mid 2010$.

2. Conversely, if $2010 = (m - 1)q$, the identity $m^q = m + (m - 1)(m + m^2 + \cdots + m^{q-1})$ expresses one power as $m + (m - 1)(q - 1) = 2011$ powers. Here the initial m denotes m copies of m^0, and the empty sum when $q = 1$ causes no problem.

3. There are $2^4 = 16$ positive divisors of $2010 = 2 \cdot 3 \cdot 5 \cdot 67$, and hence 16 values of m.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-I-Q10] | Part B (Q6-Q10)

Answer: 170

1. A square $u^2 \equiv 256 \pmod{1000}$ requires $4 \mid u$ and $u \equiv \pm 16 \pmod{125}$: the two factors $u-16, u+16$ differ by thirty-two, so only one can contain factors of five.

2. Combining gives $u \equiv \pm 16 \pmod{500}$. The positive square roots run $16, 484, 516, 984, 1016, \ldots$, with the tenth equal to 2484.

3. The tenth truncated square is $(2484^2 - 256)/1000 = (2468 \cdot 2500)/1000 = 6170$, leaving remainder 170.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2014-I-Q08] | Part B (Q6-Q10)

Answer: 937

1. The ending is an idempotent modulo 10000: $N(N-1)$ is divisible by $16 \cdot 625$. Since consecutive integers are coprime, each prime-power factor lies entirely in one of them.

2. The four Chinese-remainder choices give endings $0000, 0001, 0625, 9376$. Only the last has a nonzero thousands digit.

3. Thus the requested three-digit number is 937.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2017-I-Q09] | Part B (Q6-Q10)

Answer: 045

1. Modulo99, multiplying by100 does nothing. Hence $a_n \equiv 10 + 11 + \cdots + n = (n + 10)(n - 9)/2$. Since2 is invertible modulo99, the condition is $99 \mid (n + 10)(n - 9)$.

2. The two factors differ by19, coprime to99, so each of the prime powers9 and11 must divide a single factor. The four resulting CRT classes modulo99 are $n \equiv 9, 45, 53, 89$.

3. The least representative greater than10 is45; its sum is $55 \cdot 36/2 = 990$, divisible by99.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2017-II-Q08] | Part B (Q6-Q10)

Answer: 134

1. After multiplying by720 and removing integral contributions from $1+n$, the condition is divisibility of $F=n^6+6n^5+30n^4+120n^3+360n^2$ by720. Modulo2 and3, only n^6 can obstruct divisibility, so $6 \mid n$.

2. Write $n=6t$. The last three terms are now multiples of720, while the first two combine to $6^6t^5(t+1)$. Since $720=2^4\cdot 3^2\cdot 5$, the only remaining condition is $t\equiv 0$ or4 modulo5.

3. The range is $1\le t\le 336$. Each allowed residue occurs67 times, giving134 integers.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2020-II-Q10] | Part B (Q6-Q10)

Answer: 239

1. Put $K = n(n+1)/2$, so the sum is K^2. Modulo $d = n+5$, $2K \equiv 20$ and therefore $4K^2 \equiv 400$. The desired condition implies $d \mid 332$, with $d > 17$.

2. Thus $d = 83, 166, 332$, corresponding to $n = 78, 161, 327$. Since multiplication by 4 may lose information, test the original congruence: the respective residues $K^2 \bmod d$ are $17, 17, 100$.

3. Only $78, 161$ work, and their sum is 239.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2022-II-Q08] | Part B (Q6-Q10)

Answer: 080

1. Each triple corresponds to a consecutive block of integers. A singleton block requires a jump immediately before and after n: some member of $\{4, 5, 6\}$ divides n and some member divides $n + 1$. The endpoint $n = 1$ is not unique and adds no exception.

2. The even divisors $4, 6$ cannot divide consecutive integers on opposite sides. Thus the divisor pairs are $(4, 5), (5, 4), (6, 5), (5, 6)$. Solving modulo 60 gives their union $\{4, 5, 15, 24, 35, 44, 54, 55\}$.

3. Each of these eight residues occurs ten times in $1, \ldots, 600$, so the count is 80. Uniqueness is checked globally, including potential matches above 600.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2023-I-Q07] | Part B (Q6-Q10)

Answer: 049

1. Write the remainders as $r_2, \ldots, r_6$. For even n, distinctness forces $r_2 = 0, r_4 = 2, r_6 = 4$, then $r_3 = 1, r_5 = 3$. These are exactly $n \equiv 58 \pmod{60}$.

2. For odd n, one must have $r_2 = 1, r_4 = 3, r_6 = 5, r_3 = 2$, leaving $r_5 = 0$ or 4. The resulting classes modulo 60 are 35 and 59.

3. The classes $35, 58, 59$ have respectively $17, 16, 16$ representatives in $1, \ldots, 999$. Their sum is 49.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2024-II-Q07] | Part B (Q6-Q10)

Answer: 699

1. Let the digits be a, b, c, d and $k \equiv N \pmod 7$. Replacement conditions imply $k \equiv 6(a-1) \equiv 2(b-1) \equiv 3(c-1) \equiv d-1$. Thus $(a, b, c, d) \equiv (1-k, 1+4k, 1+5k, 1+k) \pmod 7$.

2. Substituting into $k \equiv 6a + 2b + 3c + d$ gives $k \equiv 5 + 4k$, so $k = 3 \pmod 7$. The digit residues are $(5, 6, 2, 4)$.

3. Choose the largest allowable digit in each class: $N = 5694$. Replacements yield $1694, 5194, 5614, 5691$, all divisible by 7. Hence $Q + R = 5 + 694 = 699$.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2026-I-Q08] | Part B (Q6-Q10)

Answer: 244

1. Factor $17017 = 7 \cdot 11 \cdot 13 \cdot 17$. A divisor is $7^a 11^b 13^c 17^d$, each exponent from zero to seventeen. Modulo twelve, only the parities of a, b, d matter, since the residues are $7, 11, 1, 5$.

2. Of eight parity patterns, exactly $(a, b, d) \equiv (0, 0, 1), (1, 1, 0) \pmod 2$ produce residue five. Each parity allows nine exponent choices and c has eighteen choices. Thus $N = 2 \cdot 9^3 \cdot 18 = 26244$, remainder 244.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-I-Q14] | Part C (Q11-Q15)

Answer: 030

1. If the size is n and sum S, each deletion requires $S - x \equiv 0 \pmod{n-1}$. Thus all elements are $1 \pmod{n-1}$, and $n-1 \mid 2001 = 3 \cdot 23 \cdot 29$.

2. Distinctness gives $2002 \geq 1 + (n-1)^2$, so $n - 1 \leq 44$. The largest divisor of 2001 in that range is 29, yielding $n \leq 30$.

3. Take $1, 30, 59, \ldots, 813$ (twenty-nine terms) together with 2002. All thirty terms are $1 \pmod{29}$, making their total $1 \pmod{29}$ and every deletion divisible by 29. Thus 30 is attainable.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2011-II-Q14] | Part C (Q11-Q15)

Answer: 440

1. For each modulus m, all positions of one residue class receive values from a single residue class. Every position class and every value class has $30/m$ members.

2. Two position classes cannot map to the same value class, as that would require twice as many distinct values as available. Thus the induced residue map is a permutation σ_m.

3. Conversely, any three residue permutations uniquely specify every value by the Chinese remainder theorem modulo 30, and map distinct position triples to distinct value triples.

4. The count is $2!3!5! = 1440$, leaving remainder 440.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-I-Q15] | Part C (Q11-Q15)

Answer: 332

1. Permutation of the seats requires $\gcd(a, n) = 1$. A nonzero seat difference d must also obey $ad \not\equiv d, -d \pmod{n}$, equivalent to multiplication by $a - 1$ and $a + 1$ having no nonzero kernel.

2. Thus all of $a - 1, a, a + 1$ must be coprime to n. Three consecutive integers always contain multiples of both two and three, so n can have neither prime factor.

3. Conversely, if $\gcd(n, 6) = 1$, choose $a = 3$. The consecutive product $2 \cdot 3 \cdot 4$ is coprime to n, proving sufficiency.

4. The integers in range are $6k \pm 1$, $1 \leq k \leq 166$, numbering 332.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-II-Q12] | Part C (Q11-Q15)

Answer: 958

1. The allowed residues modulo p are $3, \ldots, p-3$. There are $2, 6, 8$ choices for the three primes, so the Chinese remainder theorem gives 96 allowed classes modulo 1001.

2. Ten complete blocks, 1 through 10010, contain 960 safe numbers. Among the excess last ten numbers, only $10006, 10007$ are seven-safe; their residues modulo eleven are $7, 8$ and modulo thirteen $9, 10$, so both are fully safe.

3. Subtracting them gives 958.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-I-Q11] | Part C (Q11-Q15)

Answer: 148

1. Write $N = 1680k$. Below fourteen, only $9, 11, 13$ fail to divide 1680, so they must be the three remainder-three divisors.

2. The congruences reduce to $k \equiv 2 \pmod 3, k \equiv 10 \pmod{11}, k \equiv 1 \pmod{13}$. Combining gives $k \equiv 131 \pmod{429}$; for example 131 satisfies all, and the Chinese remainder theorem makes the class unique.

3. The least positive k is 131, a prime. The distinct primes of N are $2, 3, 5, 7, 131$, whose sum is 148.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-I-Q15] | Part C (Q11-Q15)

Answer: 272

1. Write $a = b + d, c = b + 2d$, where $0 < 2d < p$. If the original progression has difference D, then $D \equiv -d$ and $D \equiv 2d \pmod{p}$. Thus $p \mid 3d$ and $p \nmid d$, forcing $p = 3$.

2. Then $d = 1$ and $(b, a, c) = (0, 1, 2)$. Exactly the progressions $A = 1 + 3j, D = 2 + 3k$ qualify, with nonnegative integers j, k.

3. The upper bound is $C = 5 + 3j + 6k \leq 99$, or $j + 2k \leq 31$. Summing $32 - 2k$ over $k = 0, \ldots, 15$ gives 272.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-II-Q14] | Part C (Q11-Q15)

Answer: 512

1. A remainder is at most both $k-1$ and $n-2k$, since the quotient is at least two. Write $n=3m+r$, $r=0,1,2$; the choice $k=m+1$ gives remainder $m+r-2$ for this range of n.

2. For $k \geq m+1$, the bound $n-2k$ proves optimality. For smaller k, $k-1$ does so, except when $r=0, k=m$, which instead has zero remainder. Thus $F(3m)=m-2, F(3m+1)=m-1, F(3m+2)=m$.

3. Group $20,\ldots,100$ into triples $3m-1, 3m, 3m+1$ for $m=7,\ldots,33$. Each contributes $3m-4$, so the sum is $3(7+33)27/2-4\cdot 27=1512$.

4. The remainder is 512.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2019-II-Q14] | Part C (Q11-Q15)

Answer: 071

1. For each residue r modulo five, let M_r be its least representable amount. Adding fives then represents every larger amount in that residue, so the largest missing amount is $\max_r M_r - 5$. Thus the maximum residue minimum must be 96, and in particular $M_1 = 96$.

2. Using k stamps of values $n, n+1$ produces $kn + c$ for $0 \le c \le k$. According as $n \equiv 0, 1, 2, 3, 4 \pmod 5$, the least residue-one amounts are $n+1, n, 2n+2, 2n, 4n$. Setting each to ninety-six gives candidates $95, 96, 47, 48, 24$.

3. For 95 a residue-four minimum exceeds ninety-six; for 96 a residue-three minimum does; for 48 the residue-two minimum is ninety-seven. The survivors $24, 47$ have all residue minima at most ninety-six, with the required maximum. Their sum is 71.

Source / 来源

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2021-II-Q13] | Part C (Q11-Q15)

Answer: 797

1. $n = 1, 2$ fail. For $n \geq 3$, modulo 8 forces n odd and $n \equiv 5 \pmod 8$; modulo 125 the condition is $n \equiv 2^n \pmod{125}$.

2. First $n \equiv 1 \pmod 4$ yields $2^n \equiv 2 \pmod 5$, so $n \equiv 17 \pmod{20}$. Then $2^{17} \equiv 22 \pmod{25}$ implies $n \equiv 97 \pmod{100}$.

3. Finally $2^{97} \equiv 47 \pmod{125}$, giving $n \equiv 47 \pmod{125}$. Together with $n \equiv 5 \pmod 8$, this forces $n \equiv 797 \pmod{1000}$. Direct modular substitution verifies 797, so it is least.

Source / 来源